

Augmenting BBS with Conventional Signatures

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Abstract. Anonymous credential schemes such as BBS face a significant deployment barrier: currently available secure hardware such as HSMs required for eIDAS Level of Assurance High does not yet support BBS signatures or pairing-friendly curves. We address this challenge by augmenting BBS credentials with a conventional signature (such as ECDSA), where the issuer additionally signs part of the BBS signature using a conventional signature private key that can be secured in widely available HSMs. We present two disclosure protocols for augmented BBS credentials: an unlinkable protocol that uses zkSNARKs to hide it (though the practical viability of the latter requires further study), and a simple linkable protocol that directly reveals the conventional signature. For the linkable protocol, we argue that the loss of unlinkability is acceptable for high-assurance use cases where disclosed attributes already uniquely identify the user. We prove that augmented BBS credentials are existentially unforgeable under chosen message attacks, with security depending solely on the conventional signature private key rather than the BBS private key. This approach provides a practical migration path to full BBS deployment while maintaining several key BBS advantages.

1 Introduction

A major deployment barrier for BBS [BBS04; ASM06; CDL16; TZ23] is that currently available secure hardware (e.g., HSMs) to secure the issuer private key does not yet support the BBS signature scheme, nor pairing-friendly curves on which BBS operates. Since this is expected to be a requirement for [eIDAS] Level of Assurance (LoA) High, this means that without additional measures BBS would not be able to achieve eIDAS LoA High.

Since HSM support for securing private keys of conventional signature schemes (i.e., signature schemes signing a single message that don't support unlinkability, such as ECDSA) is widely available, in this paper we introduce a way of augmenting a BBS credential with such a conventional signature, where the issuer signs part of the BBS signature with a conventional signature key secured in an HSM. Essentially, we augment the BBS signature with a conventional signature.

We present two disclosure protocols for augmented BBS credentials. The first is a sketch of an unlinkable protocol that uses zkSNARKs to hide the conventional signature, though further study is required to establish the specifics of the zkSNARK as well as the performance of this approach. The second is a

simple linkable protocol that includes the conventional signature directly in the disclosure, sacrificing unlinkability.

We argue that despite the loss of unlinkability, the linkable disclosure protocol can still be of value:

- In use cases where the additional security of the conventional signature is not required, it can simply be ignored while the BBS credential is disclosed normally. For such use cases unlinkability would be preserved.
- It is to be expected that the majority of LoA High use cases will require the user to be uniquely identified through the disclosed attributes. In such use cases, even if the disclosure protocol would be unlinkable, that unlinkability would not offer added value. For these cases, we can safely omit the unlinkability offered by the BBS disclosure protocol by removing some of the blinding operations, which makes it possible to include the conventional signature in the disclosure.

We show in this paper that the unforgeability of augmented BBS credentials depends solely on the conventional signature private key and not on the BBS private key, so that it is no longer an issue that the latter currently cannot be HSM-secured. In addition, we show that the modifications that we make to the BBS disclosure protocol in order to enable including the issuer's conventional signature do not harm unforgeability.

While linkability is a limitation, practical deployment considerations can justify this trade-off. For use cases where disclosed attributes don't uniquely identify users and don't require the additional security of the conventional signature, the BBS unlinkability properties can still be used by simply not using the conventional signature. Moreover, despite linkability, augmented BBS credentials retain many of the advantages that BBS offers over conventional signature-based credentials such as [ISO-18013-5] mdocs and [SD-JWT]:

- For the undisclosed attributes in an augmented BBS credential, predicate proofs such as range proofs and set membership remain available, as well as other techniques such as for example attribute-based encryption.
- Using the standard mechanism of having the issuer(s) blindly issue a single random number as the first attribute of all of the user's credentials, the user can prove to the verifier that two (augmented) BBS credentials come from a single wallet by proving that the first attribute of all disclosed credentials have the same value (called Cryptographic Binding of Attestations, or Proofs of Association, in EUDI context).¹
- Unlike conventional signature-based credentials such as mdoc and SD-JWT, when given a disclosure transcript verification of the disclosed attributes always requires the verifier's nonce, irrevocably binding attributes to the disclosure session.

¹ In credentials such as mdocs or SD-JWT, one could also include a random number as the first attribute of all of the user's credential for this purpose, but not without all issuers getting to see its value, as well as all verifiers whenever it is used.

As such, even when disclosed linkably, augmented BBS credentials offer everything that credentials such as ISO 18013-5 mdoc and SD-JWT offer, and more: not only the above benefits in use cases where the additional security of the conventional signature is required, but also unlinkability in use cases where it is not.

Finally, augmented BBS credentials provide a practical deployment path that enables organizations to build BBS infrastructure today rather than waiting for HSM support for BBS and pairing-friendly curves. Once such HSM support becomes available, the transition to fully unlinkable BBS credentials requires minimal protocol changes, wallet software updates, and system migrations: the issuer simply stops adding the conventional signature. This forward compatibility means that the entire ecosystem can be deployed now, with unlinkability for LoA High use cases gained automatically as a future upgrade when secure hardware technology catches up.

2 Augmented BBS credentials

We start with establishing some notation by describing the BBS signature scheme, as described in [TZ23]. Let $\mathbb{Z}_p = \mathbb{Z}/p\mathbb{Z}$, and let $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ be a Type 3 pairing-friendly curve of prime order p . A BBS signature over messages $(m_1, \dots, m_n) \in \mathbb{Z}_p^n$ is a tuple $(A, e) \in \mathbb{G}_1 \times \mathbb{Z}_p$ such that

$$A^{x+e} = g \prod_i h_i^{m_i}$$

in which

- The elements $g, h_1, \dots, h_n \in \mathbb{G}_1$ are generators.
- The number $x \in \mathbb{Z}_p$ is the issuer's BBS private key.

A BBS credential is a tuple $(A, e, (m_1, \dots, m_n))$ where (A, e) is valid over the messages $(m_1, \dots, m_n)$. The disclosure process of such a credential works as follows (following Appendix B of [TZ23]). Let $\mathcal{C}$ be the set of hidden attributes, and $\mathcal{D}$ the set of disclosed attributes.

1. The user chooses random scalars $r_1, r_2 \in \mathbb{Z}_p$, and computes

$$\bar{A} = A^{r_1/r_2}, \quad D = \left(g \prod_{i=1}^n h_i^{m_i} \right)^{1/r_2}, \quad \bar{B} = D^{r_1} \bar{A}^{-e}$$

2. The user sends $\bar{A}, \bar{B}$ and D to the verifier, and performs the following zero-knowledge proof:

$$\text{PK} \left\{ (e, r_1, r_2, (m_i)_{i \in \mathcal{C}}) \left| \bar{B} = D^{r_1} \bar{A}^{-e}, g \prod_{i \in \mathcal{D}} h_i^{m_i} = D^{r_2} \prod_{i \in \mathcal{C}} h_i^{-m_i} \right. \right\}$$

The verifier accepts if the proof is valid, and if

$$e(\bar{A}, X) = e(\bar{B}, g)$$

where $X \in \mathbb{G}_2$ is the issuer's public key.

Definition 1. An *augmented BBS credential* is a tuple $(A, e, \sigma, (m_1, \dots, m_n))$ such that

- (A, e) is a BBS signature over $(m_1, \dots, m_n)$.
- σ is a conventional signature (such as ECDSA) over A .

In a disclosure protocol in which σ is sent to the RP, the value A that it signs will also have to be sent for σ to be of any use. Therefore, in such a disclosure protocol A must not be randomized into $\bar{A} = A^{r_1/r_2}$ the way it is above. Removing that from the disclosure protocol, i.e., setting $r_1 = r_2 = 1$, results in the following.

Definition 2. An augmented BBS credential can *linkably* be disclosed as follows.

1. The user computes

$$D = g \prod_{i=1}^n h_i^{m_i}, \quad B = DA^{-e}$$

2. The user sends A, B, D and the conventional signature σ , and performs the following zero-knowledge proof:

$$\text{PK} \left\{ (e, (m_i)_{i \in \mathcal{C}}) \left| B = DA^{-e}, g \prod_{i \in \mathcal{D}} h_i^{m_i} = D \prod_{i \in \mathcal{C}} h_i^{-m_i} \right. \right\}$$

The verifier accepts if (1) the proof is valid, (2) σ is a valid conventional signature over A , and (3) if

$$e(A, X) = e(B, g)$$

where $X \in \mathbb{G}_2$ is the issuer's public key.

Before disclosure, the verifier will have to decide whether or not it requires the conventional signature augmentation of a BBS credential, and consequentially the above linkable disclosure protocol. This decision will depend on the specifics of the use case, in particular the required LoA and whether or not the attributes uniquely identify the user. As such, this is outside of the scope of this paper, but we do note the following.

- If the system has credential metadata that defines the structure and semantics of the credential and the contained attributes, then for each attribute that metadata could include whether or not the attribute is user identifying or not. For example, a social security number or membership number is identifying, while an “older than 18” attribute is not. When identifying attributes are disclosed, the verifier is free to choose the linkable disclosure protocol without introducing privacy consequences.
- The disclosure protocol will have to indicate to the wallet if the linkable disclosure protocol is required by the verifier, or if the normal, unlinkable BBS disclosure protocol suffices.

3 Security

An augmented BBS signature can be considered as a signature made with the conventional signature private key over the messages $(m_1, \dots, m_n)$. In the theorem below, we show that these signatures are existentially unforgeable under chosen message attacks (EUF-CMA) [GMR88].

Theorem 1. *Assuming hardness of solving discrete logarithms, and unforgeability of the conventional signature scheme, no probabilistic polynomial-time algorithm can win the EUF-CMA unforgeability game for augmented BBS credentials with non-negligible probability, regardless of the secrecy of the BBS private key x .*

Proof. Suppose there exists an adversary $\mathcal{A}$ that with non-negligible probability can forge an augmented BBS credential over $(m_1, \dots, m_n)$, without it having queried a signature on those messages in the EUF-CMA unforgeability game. By assumption, at the end of the unforgeability game the adversary outputs a valid augmented BBS credential $(A, e, \sigma, (m_1, \dots, m_n))$. This means that the following holds:

$$A^{x+e} = g \prod_i h_i^{m_i}$$

Since the conventional signature scheme is assumed unforgeable, the signature σ over A must have been created by the challenger in one of the queries performed by the adversary to the challenger. In that query, the adversary received a BBS signature (A, e') over attributes $(m'_1, \dots, m'_n)$ of its choice, satisfying

$$A^{x+e'} = g \prod_i h_i^{m'_i}$$

Rearranging the two equations and combining them, we find

$$A^{-e} \prod_i h_i^{m_i} = g^{-1} A^x = A^{-e'} \prod_i h_i^{m'_i}$$

Dividing the left-hand side by the right-hand side results in the following DL-representation of 1:

$$A^{e'-e} \prod_i h_i^{m_i - m'_i} = 1$$

Since the adversary won the unforgeability game by assumption we must have $(m_i)_i \neq (m'_i)_i$, so that this DL-representation of 1 is nontrivial. Since being able to create nontrivial DL-representations of 1 is equivalent to being able to break the discrete logarithm problem, this contradicts the discrete logarithm assumption.

Notice that nowhere in this proof did we use anything related to the unforgeability of BBS or the secrecy of its private key. We only used the validity equation of the BBS signature scheme, the unforgeability of the conventional signature scheme, and the hardness of the discrete logarithm. Indeed, even if we

had given the adversary knowledge of the value x , for example by including it in the public key, the argument above would still hold. Therefore, whether or not the BBS private key x is kept secret does not matter. $\square$

It is important that the disclosure protocol should have the property that the verifier cannot disclose the attributes that it receives from the user towards another verifier (i.e., impersonate the user), even in use cases where the user discloses all of the attributes from the credential. Disclosure protocols that consist in their entirety of a zero-knowledge proof of a valid signature automatically have this property due to the zero-knowledge property, but the linkable disclosure protocol from Definition 2 is not a zero-knowledge proof of an augmented BBS credential, since part of the signature is not hidden behind a zero-knowledge proof but instead sent directly to the verifier.

For that reason, the definition below presents a model of what it means for a disclosure protocol to be unforgeable, in the form of an unforgeability game played between the adversary and a challenger. This game is modeled on the EUF-CMA game, with the following changes:

- In the query phase, in order to model the situation where the adversary might act as the verifier to try to learn information about credentials that are disclosed to it, we endow it with a query for executing the disclosure protocol on attributes of its choice.
- The final phase, in which normally the adversary must show its forged signature, is replaced with the disclosure protocol. In order to win, the adversary must disclose credentials that were not issued to it.

Definition 3. The unforgeability game for a disclosure protocol between a challenger and an adversary $\mathcal{A}$ is defined as follows.

Setup The adversary decides on the number of attributes $n \geq 1$ that each credential will have, and sends n to the challenger. The challenger then runs the key generation algorithm from the credential scheme and sends the resulting public key to the adversary.

Queries The adversary $\mathcal{A}$ can make the following queries to the challenger.

Sign $(m_{1,j}, \dots, m_{n,j})$ The adversary requests a signature on the attributes $(m_{1,j}, \dots, m_{n,j})$, which it may choose adaptively, and receives the signature from the challenger.

Disclose $(\mathcal{D}, m_1, \dots, m_n)$ The challenger creates a credential with the specified attributes $m_1, \dots, m_n$, and engages in the Disclose protocol with the adversary, acting as the user and taking $\mathcal{D}$ as disclosure set, while the adversary acts as the verifier.

Challenge The challenger, now acting as the verifier, and the adversary, acting as the user, engage in the Disclose protocol. The adversary chooses a nonempty disclosure set $\mathcal{D}$, and if it manages to make the verifier accept then it wins if the following holds: if $(m_i)_{i \in \mathcal{D}}$ are the disclosed attributes, then there must be no single credential issued in an Sign query containing all of the disclosed attributes $(m_i)_{i \in \mathcal{D}}$.

A disclosure protocol is said to be unforgeable if no adversary can win this game with non-negligible probability.

Theorem 2. *The linkable disclosure protocol for augmented BBS credentials from Definition 2 is unforgeable, assuming hardness of computing discrete logarithms and that the conventional signature scheme is EUF-CMA unforgeable, and regardless of the secrecy of the BBS private key x .*

Proof. Suppose there is an adversary $\mathcal{A}$ that can win the unforgeability game of the disclosure protocol from Definition 2. We construct a challenger $\mathcal{C}$ that plays the unforgeability game with $\mathcal{A}$. This challenger $\mathcal{C}$ itself simultaneously plays the EUF-CMA unforgeability game for the conventional signature scheme, acting as the adversary; let $\mathcal{C}_{\text{sig}}$ be the challenger in that EUF-CMA game.

In the setup phase, $\mathcal{C}$ takes a random generator $\bar{g} \in \mathbb{G}_1$, takes random $a, b_i \in \mathbb{Z}_p$, and sets $g = \bar{g}^a$, $h_i = \bar{g}^{b_i}$. Next it generates a BBS private key $x \in \mathbb{Z}_p$, sets the BBS public key $X = g^x$, and sends the conventional signature public key, $g, (h_i)_i, X$ to $\mathcal{A}$.

The challenger $\mathcal{C}$ responds to queries of $\mathcal{A}$ as follows.

Sign($m_{1,j}, \dots, m_{n,j}$) Choose a random $e \in \mathbb{Z}_p$, set $A = (g \prod_i h_i^{m_{i,j}})^{1/(x+e)}$, and query $\mathcal{C}_{\text{sig}}$ for a conventional signature σ on A . Return (A, e, σ) to $\mathcal{A}$.

Disclose($\mathcal{D}, m_1, \dots, m_n$) Suppose we wish to compute the discrete logarithm of some element $A \in \mathbb{G}_1$. Query $\mathcal{C}_{\text{sig}}$ for a conventional signature σ on A , and perform the disclosure protocol with $\mathcal{A}$ by using the simulator for the zero-knowledge proof.

In the challenge phase of the game, $\mathcal{C}$ uses the extractor for the zero-knowledge proof performed by $\mathcal{A}$ to extract from it numbers $e, (m_i)_{i \in \mathcal{D}}$ such that $(A, e, \sigma, (m_1, \dots, m_n))$ is a valid augmented BBS credential. (Note that while $\mathcal{D}$ may be empty, $\mathcal{A}$ always proves knowledge of the number e .)

Now we distinguish three possibilities.

- A is new, in the sense that it was not used in one of the queries. Then $\mathcal{C}$ can win the conventional signature unforgeability game by forwarding (σ, A) to $\mathcal{C}_{\text{sig}}$, the probability of which is negligible by assumption.
- A is not new, and was used in a **Sign** query. For this case the argument from the proof of Theorem 1 can be used to show that this, too, can happen at most with negligible probability.
- A is not new, and was used in a **Disclose** query. Then we must have

$$A = \left(g \prod_i h_i^{m_i} \right)^{1/(x+e)} = \bar{g}^{(a+b_i m_i)/(x+e)}$$

This means that we have computed the discrete logarithm of the element A with respect to $\bar{g}$. This too can be the case with at most negligible probability.

Since each of the three possibilities can happen only with negligible probability, we conclude that $\mathcal{A}$ can win the unforgeability game of the disclosure protocol with at most a negligible probability. $\square$

This proof depends on the fact that even though the element A is sent to the verifier, the value e remains hidden from the verifier and is only proven in zero-knowledge. This means that even if all the attributes $(m_1, \dots, m_n)$ were to be disclosed to the verifier, it still does not gain sufficient information to replay what it receives to another verifier: without knowledge of e it cannot construct a valid zero-knowledge proof towards another verifier. Therefore, replay protection is maintained regardless of how many attributes are disclosed.

4 Unlinkably disclosing augmented BBS credentials

In this section, we sketch an unlinkable disclosure protocol for augmented BBS credentials that uses zkSNARKs (i.e., circuit-based zero-knowledge proofs) to hide the conventional signature. We note that for elliptic curve based conventional signature schemes such as EdDSA or ECDSA, such a system would require elliptic curve operations on two distinct curves: the one that the conventional signature scheme uses, and the BBS curve. This is expected to complicate the required circuits as well as the software to realize it. Further study is required to examine the impact of this on the performance of this solution when compared with alternatives such as Longfellow [Fs24]. Nevertheless, we present our protocol below.

Definition 4. An augmented BBS credential can *unlinkably* be disclosed as follows.

1. The wallet performs the (normal) BBS disclosure process as described in Section 2.
2. Next, it sets $r = r_2/r_1$ (where r_1, r_2 are as described in Section 2), and then uses a circuit-based zero-knowledge proof to prove that it knows a conventional signature σ and a number $r \in \mathbb{Z}_p$ such that σ is a conventional signature over $\bar{A}^r$ that verifies against the issuer's public key.

The verifier accepts if the BBS disclosure and the circuit-based zero-knowledge proof are both valid.

It should be noted that this disclosure protocol, while consisting of two zero-knowledge protocols, is not itself a zero-knowledge proof of having an augmented BBS credential, because while for honest users $r = r_2/r_1$ holds, the proof does nothing to convince the verifier that this is indeed the case. This means that it is not guaranteed by the proof that the conventionally signed $\bar{A}^r$ equals $\bar{A}^{r_2/r_1} = A$, which is necessary for (A, e, σ) to be an augmented BBS signature.

Nevertheless, it is possible to show unforgeability of this disclosure protocol in the sense of Definition 3, along the lines of the previous proofs. However, when compared with the proof of Theorem 1, we have to be careful to take into account r versus r_1 and r_2 .

Theorem 3. Let $(A, e, \sigma, (m_1, \dots, m_n))$ be an augmented BBS credential. Assuming hardness of solving discrete logarithms and EUF-CMA unforgeability

of the conventional signature scheme, the unlinkable disclosure protocol of augmented BBS credentials is unforgeable, regardless of the secrecy of the BBS private key x .

Proof. Suppose there exists an adversary $\mathcal{A}$ that with non-negligible probability convinces the verifier using the unlinkable disclosure protocol that it has an augmented BBS credential over $(m_1, \dots, m_n)$, without it having queried a signature on those messages. We construct a challenger exactly as we did in the proof of Theorem 2, except that in the challenge phase, it acts as follows.

Given the extractor algorithms for the circuit-based zero-knowledge proof and for the zero-knowledge proof that is part of the BBS disclosure process, the challenger extracts the following from the adversary $\mathcal{A}$:

- A number r and a conventional signature σ such that σ is a valid conventional signature over $\bar{A}^r$.
- Numbers $e, r_1, r_2, (m_i)_i$ such that the tuple $(\bar{A}^{r_2/r_1}, e, (m_1, \dots, m_n))$ is a valid BBS credential. This means that the following holds:

$$\bar{A}^{(r_2/r_1)(x+e)} = g \prod_i h_i^{m_i}$$

Set $A' = \bar{A}^r$. Using the extracted information, we distinguish the same three cases as in the proof of Theorem 2:

- A' is new, in the sense that it was not used in one of the queries. Then our challenger can win the conventional signature unforgeability game by forwarding (σ, A') to the challenger of the conventional signature scheme, the probability of which is negligible by assumption.
- A' is not new, and was used in a Disclose query. Then we must have

$$A' = \left(g \prod_i h_i^{m_i} \right)^{r(r_1/r_2)/(x+e)} = \bar{g}^{r(r_1/r_2)(a+b_i m_i)/(x+e)}$$

This means that we have computed the discrete logarithm of the element A' that was chosen in the Disclose query with respect to $\bar{g}$. This too can be the case with at most negligible probability.

- A' is not new, and was used in a Sign query. In that query, the adversary received a BBS signature (A', e') over attributes $(m'_1, \dots, m'_n)$ of its choice, satisfying (now writing $\bar{A}^r$ again instead of A'):

$$\bar{A}^{r(x+e')} = g \prod_i h_i^{m'_i}$$

Rearranging the two equations and combining them, we find

$$\bar{A}^{-e} g^{r_1/r_2} \prod_i h_i^{m_i r_1/r_2} = \bar{A}^x = \bar{A}^{-e'} g^{1/r} \prod_i h_i^{m'_i/r}$$

Dividing the left-hand side by the right-hand side results in the following DL-representation of 1:

$$\bar{A}^{e'-e} g^{r_1/r_2-r} \prod_i h_i^{m_i r_1/r_2 - m'_i/r} = 1$$

If $r_1/r_2 \neq r$, then this is a nontrivial DL-representation of 1. If $r_1/r_2 = r$, then the exponents of h_i become $m_i - m'_i$, and since $(m_i)_i \neq (m'_i)_i$ as the adversary won the unforgeability game, this means that the above expression is still a nontrivial DL-representation of 1. $\square$

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